

FARI

Framework Validation Report

Process Validation and Observed Framework Behavior

Release source: FARI-PROCESS-VALIDATION.md

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This is the versioned release edition of the FARI documentation bundle. Canonical working sources for this release are stored under the same version folder.

Purpose

This document records what the first four case reports revealed about the FARI process and which changes were made before treating the framework as stable.

Cases Used

Case	Evidence Type	Report Outcome
01_Lib.md	Raw fuzzing and AddressSanitizer logs with short analyst conclusion	Does Not Meet; confirmed technical claim failure
02_USB.md	Test concept, script, reproduction plan, expected observations, analyst conclusion	Inconclusive; extended investigation required
03_USB.md	Source excerpt, malformed USB telecommand screenshot, and impact assertions	Inconclusive; delivery demonstrated but effect unresolved
04_Spacecanbus.md	Bus injection/replay commands and monitor screenshots	Does Not Meet in lab monitoring scope; broader effects unresolved

What Worked

1. **Method agnosticism worked.** FARI accepted raw fuzzing logs and a hardware-interface test concept without prescribing either technical method.
2. **SPARTA added useful context.** The reports corrected or refined mappings without changing the supplied technical evidence.
3. **Risk, confidence, and coverage remained separate.** Confirmed technical defects did not automatically become confirmed mission-level denial of service.
4. **Executive decisions were possible with incomplete evidence.** Both reports identify safe next actions while remaining provisional.
5. **A portfolio conclusion remained simple as cases increased.** Four investigation reports totalize into one result without averaging technical findings or inventing a numeric score.

Problems Revealed

PV-001: Evidence Producer and Report Author Were Not Explicitly Separate

The original process assumed the assessment team would both execute testing and write the final report. The intended operating model requires the auditor to produce evidence and another role to create the FARI report.

Change made: Added explicit Evidence Producer, Report Author, Mission Owner, and Risk Authority roles.

PV-002: Raw Source Material Can Support Different Strengths of Conclusion

The library case supports a confirmed memory-safety defect, but does not support confirmed external reachability or mission-level denial of service.

Change made: Added separate evidence-sufficiency decisions for technical condition, reachability, and mission consequence.

PV-003: Missing Context Must Be Added Without Becoming Fiction

Neither case supplied complete mission objectives, architecture, ownership, authorization, or consequence definitions. Reports still need enough context to be useful.

Change made: Added evidence dispositions that distinguish:

- Evidence-backed fact.
- Evidence-producer assertion.
- Report-author inference.
- Assumption.
- Missing information.

PV-004: Contradictions Need First-Class Treatment

The USB case contains APID and payload-format contradictions. Quietly selecting one value would create an unreliable report.

Change made: Added contradiction records and required follow-up evidence.

PV-005: Finding State and Report Maturity Are Different

A provisional report may contain a confirmed technical finding, while a final report may still contain accepted risks.

Change made: Added Draft, Provisional, Final, and Superseded report-maturity states.

PV-006: Supplied SPARTA Mappings Require Validation

Both cases proposed EX-0005.01 Design Flaws, but the supplied evidence is more accurately related to software code flaws and command behavior. Mapping by title alone would have reduced report quality.

Change made: Retained the rule that mappings require rationale and added mapping corrections to the case reports.

PV-007: FARI Must Not Create Work for an External Evidence Producer

The initial process still expected a structured evidence package and method-run information from the evidence producer. An external auditor may have an independent contract, methodology, and report format and should not need FARI context.

Change made: Source material may now arrive in any format. All FARI normalization, contextual completion, mapping, conclusion, and reporting duties belong to the Report Author. The former evidence-package template was replaced with an internal source-material intake record.

PV-008: Every Investigation Needs an Executive Conclusion

Finding state, report maturity, risk, confidence, and assurance did not provide one obvious answer to "how did this investigation end?"

Change made: Added the mandatory FARI Result Card with Conclusion, Confidence, Required Action, Priority, Scope Boundary, and Rationale. The library case concludes `Does Not Meet`; the USB case concludes `Inconclusive`.

PV-009: Inconclusive Is a Valid but Actionable Outcome

An investigation may be unable to prove or disprove a claim because of scope, contradictions, or missing observations.

Change made: `Inconclusive` is now a first-class FARI Conclusion and must create an investigation-extension or retest action.

PV-010: Multiple Reports Need One Consolidated Assessment

Individual reports do not provide a practical executive view for an infrastructure-wide assessment.

Change made: Multi-investigation assessments now require one consolidated assessment report. Gating claims determine the overall conclusion without averaging or creating a numeric security score.

PV-011: A Generic Auditor Form Is Useful Only When It Remains Optional

The supplied cases use a recurring technical structure: scenario, access, environment, tooling, execution, finding, and mappings. That structure is useful for ingestion, but requiring it would contradict the no-external-auditor-burden principle.

Change made: Added an Optional Technical Finding Submission Form using framework-neutral language. An auditor may use it when convenient, while the Report Author remains responsible for all FARI-specific work.

PV-012: A Screenshot Can Prove an Action Without Proving Its Cause or Impact

The new USB screenshot proves that a malformed packet was constructed and sent, but a timeout alone does not prove memory corruption or persistent denial of service. The SpaceCAN screenshots show injection, replay, and changed displayed values, but do not prove physical actuator behavior.

Change made: Evidence reviews must state exactly what each artifact proves and must not use screenshots as automatic proof of causal or mission-level impact.

PV-013: One Investigation Can Contain Confirmed and Candidate Findings

The SpaceCAN case supports a demonstrated spoofing/display-trust failure while the replay-specific receiver effect remains only partially evidenced.

Change made: The investigation conclusion is derived from its gating claim, while each finding keeps its own state and evidence boundary.

PV-014: Repeated Generic Finding Text Is a Source-Material Quality Signal

Both new cases contain a copied denial-of-service statement that does not match their evidence. The repetition is not harmless formatting; it can create a false conclusion if copied into an executive report.

Change made: The intake review explicitly identifies copied or unrelated finding text as a contradiction and excludes it from the conclusion.

PV-015: SPARTA Mappings Need Technique-Level Precision

The SpaceCAN source material mapped replay to bus spoofing and uplink eavesdropping. The more precise replay mapping is EX-0001.02 Bus Traffic Replay; uplink interception is not demonstrated by a local replay file.

Change made: FARI continues to preserve proposed mappings as source material, but the Report Author validates, corrects, or rejects them with rationale.

Current FARI Reporting Workflow

Diagram source is available in the canonical Markdown edition.

Remaining Framework Validation Gaps

The four pilots validate intake, contradictory evidence, confirmed and inconclusive outcomes, SPARTA correction, and consolidation. They do not yet validate:

- A Meet s investigation supported by positive control evidence.
- A remediation retest that supersedes an earlier result.
- Mission-owner confirmation of consequence, acceptance criteria, and residual risk.
- Multiple independent evidence producers in one investigation.
- Evidence integrity records, revisions, hashes, and chain of custody.
- A Final report maturity transition.
- A full manual-to-web round trip using the canonical schema.

Recommended Next Validation

Use FARI next on a case or portfolio that contains:

- Multiple evidence producers or methods.
- At least one implemented control whose effectiveness can be credited.
- Mission-owner confirmation of consequence and residual risk.
- A remediation retest.
- Source-material revisions and integrity metadata.

That pilot will validate the transition from a Provisional report to a Final FARI report.